

1D CNN — Architecture Variant Comparison

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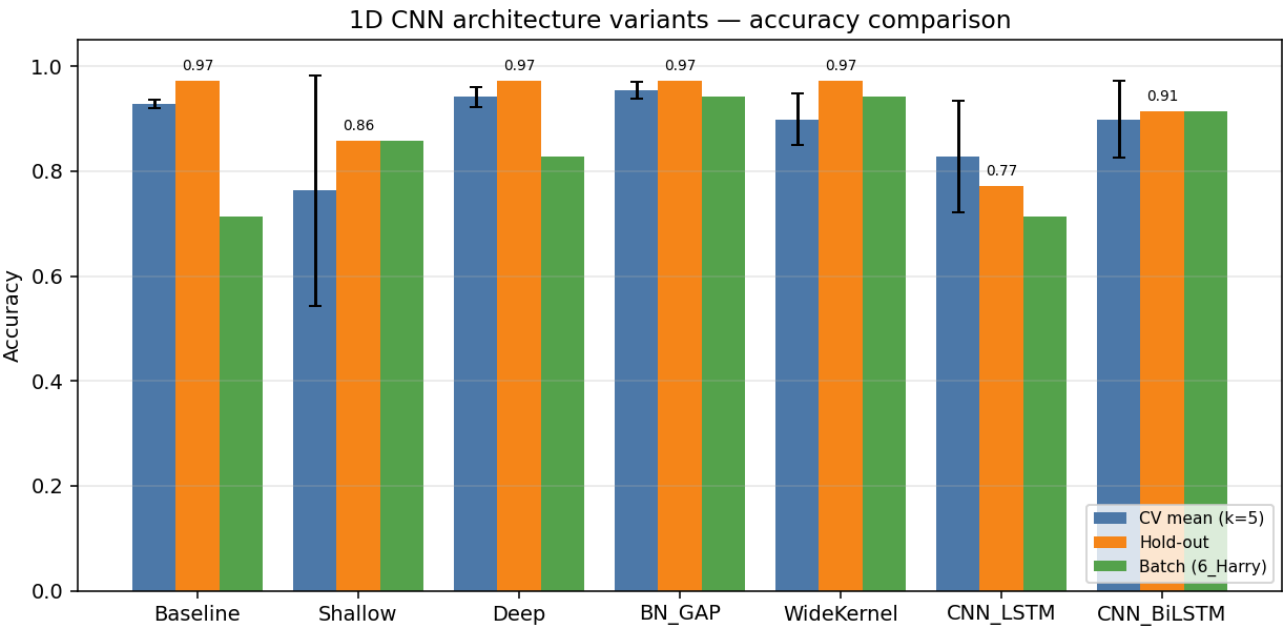
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Total trials	341 (train pool 306, hold-out 35)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

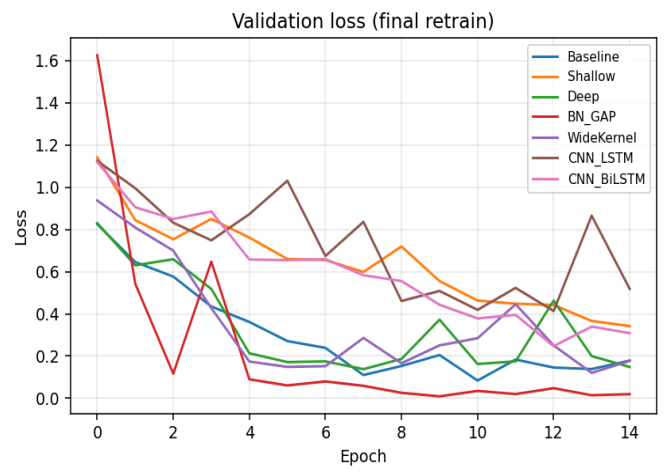
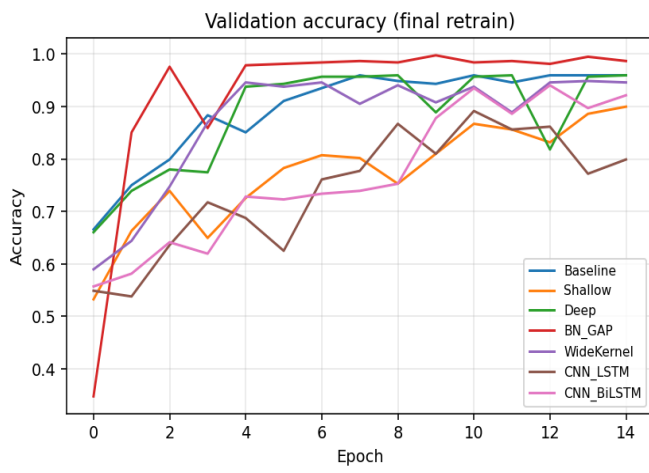
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9281	0.0082	0.9714	0.1913	0.7143	25/35
Shallow	72,487	0.7628	0.2198	0.8571	0.4084	0.8571	30/35
Deep	204,007	0.9412	0.0194	0.9714	0.1213	0.8286	29/35
BN_GAP	57,319	0.9543	0.0158	0.9714	0.1056	0.9429	33/35
WideKernel	131,687	0.8986	0.0492	0.9714	0.0941	0.9429	33/35
CNN_LSTM	68,455	0.8274	0.1060	0.7714	0.5193	0.7143	25/35
CNN_BiLSTM	105,575	0.8986	0.0729	0.9143	0.2194	0.9143	32/35

Best on hold-out: **Baseline** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9281 ± 0.0082
Hold-out test acc / loss	0.9714 · 0.1913
Batch-test acc	0.7143 (25/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9355	0.9355
2	1470	61	0.9180	0.9508	0.9180
3	1470	61	0.9344	0.9836	0.9344
4	1470	61	0.9180	0.9508	0.9180
5	1470	61	0.9344	0.9344	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	1	5	0.200
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	5	5	1.000
Overall	25	35	0.714

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.7628 ± 0.2198
Hold-out test acc / loss	0.8571 · 0.4084
Batch-test acc	0.8571 (30/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.3387	0.3387	0.3387
2	1470	61	0.8525	0.8525	0.8525
3	1470	61	0.9344	0.9672	0.9344
4	1470	61	0.7705	0.9180	0.7705
5	1470	61	0.9180	0.9180	0.9180

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	0.800	0.889	5
Double_Nothing	0.800	0.800	0.800	5
Double_Pistol_Recoil	1.000	0.800	0.889	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.600	0.750	5
macro avg	0.908	0.857	0.863	35
weighted avg	0.908	0.857	0.863	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	4	5	0.800
Overall	30	35	0.857

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9412 ± 0.0194
Hold-out test acc / loss	0.9714 · 0.1213
Batch-test acc	0.8286 (29/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9194	0.9194
2	1470	61	0.9180	0.9508	0.9180
3	1470	61	0.9508	0.9836	0.9508
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.9508	0.9672	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	0	5	0.000
Drum_Roll	5	5	1.000
Overall	29	35	0.829

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9543 ± 0.0158
Hold-out test acc / loss	0.9714 · 0.1056
Batch-test acc	0.9429 (33/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9516	0.9355
2	1470	61	0.9508	0.9836	0.9508
3	1470	61	0.9836	1.0000	0.9836
4	1470	61	0.9508	0.9672	0.9508
5	1470	61	0.9508	0.9836	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	4	5	0.800
Drum_Roll	5	5	1.000
Overall	33	35	0.943

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8986 ± 0.0492
Hold-out test acc / loss	0.9714 · 0.0941
Batch-test acc	0.9429 (33/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9355	0.9355
2	1470	61	0.9344	0.9344	0.9344
3	1470	61	0.9180	0.9508	0.9180
4	1470	61	0.8033	0.8689	0.8033
5	1470	61	0.9016	0.9344	0.9016

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	4	5	0.800
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	33	35	0.943

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8274 ± 0.1060
Hold-out test acc / loss	0.7714 · 0.5193
Batch-test acc	0.7143 (25/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.6452	0.7903	0.6452
2	1470	61	0.8525	0.9016	0.8525
3	1470	61	0.9508	0.9508	0.9508
4	1470	61	0.9016	0.9016	0.9016
5	1470	61	0.7869	0.8689	0.7869

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.571	0.800	0.667	5
Double_Pistol_Recoil	1.000	0.400	0.571	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.500	1.000	0.667	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.200	0.333	5
macro avg	0.867	0.771	0.748	35
weighted avg	0.867	0.771	0.748	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	1	5	0.200
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	25	35	0.714

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.8986 ± 0.0729
Hold-out test acc / loss	0.9143 · 0.2194
Batch-test acc	0.9143 (32/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9355	0.9194
2	1470	61	0.9344	0.9344	0.9344
3	1470	61	0.9344	0.9836	0.9344
4	1470	61	0.7541	0.8197	0.7541
5	1470	61	0.9508	0.9508	0.9508

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.400	0.571	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	0.714	1.000	0.833	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.935	0.914	0.902	35
weighted avg	0.935	0.914	0.902	35

Batch test per class (6_Harry)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	3	5	0.600
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	35	0.914